

Boxing Gym CRM Platform

Product Strategy & Engineering Infrastructure Report

1. The Business Case & Revenue Model

Building a highly targeted SaaS for boxing gyms represents a lucrative niche. Revenue generation relies on a multi-pronged approach rather than just software subscriptions.

- **Subscription Tiers:** Standard monthly licensing.
 - *Starter (\$75-\$99/mo):* Basic scheduling and billing for small independent clubs.
 - *Growth (\$150-\$199/mo):* Automated follow-ups, SMS, revenue reporting.
 - *Scale (\$250+/mo):* Branded mobile apps, advanced APIs, priority support.
- **Payment Processing Margins:** Integrating with a payment gateway to take a fractional percentage (e.g., 0.5%) on gym transactions. At 20 gyms processing \$15k/month, this adds ~\$1,500/mo in pure passive revenue.
- **Value-Added Services:** Setup fees for "done-for-you" migrations and white-glove onboarding.

2. Competitive Advantage: Niche vs. Generic

To win against massive incumbents (Mindbody, ABC Fitness), the product must abandon generic fitness workflows in favor of boxing-specific operations.

- **Streamlined UI:** Focus entirely on boxing operations (Classes, Pad Work, Fight Camp) instead of irrelevant features like yoga mat rentals.
- **Specialized Workflows:**
 - *Grade/Rank Tracking:* Systems to manage belt progression or fighter levels.
 - *Pad-Work Scheduling:* Handling complex 1:1 sessions overlaying standard group classes.
 - *Fight Camp Management:* Dedicated modules for tracking sparring frequency, weight cuts, and training load for competing fighters.
- **White-Glove Migration:** Removing the friction of switching by offering 100% managed data migration from their old software.

3. Transitioning to a Serious SaaS Product

Moving from a conceptual demo to a production-ready application requires rigorous engineering and operational shifts.

Phase 1: Validation

Conduct "smoke tests" with 10-20 local gym owners. Validate that the tool solves a hair-on-fire problem (e.g., tracking manual payments, identifying churn) before scaling development. Secure pre-sales to prove financial viability.

Phase 2: Technical Professionalism & Infrastructure

- **Security & Compliance:** Implementation of encryption (data at rest/transit), secure authentication (MFA), and automated daily database backups.
- **Payment Infrastructure:** Strictly integrating with PCI-compliant providers like Stripe or Square. Never build custom payment tokenization.
- **Modular Architecture:** Transitioning to scalable cloud infrastructure (e.g., AWS, Vercel, PostgreSQL) with decoupled services for Billing, Scheduling, and Analytics to ensure high availability.

Phase 3: Operational Strategy

Develop a "sticky" platform that gym owners cannot operate without. This involves automated retention engines (e.g., SMS check-ins for absent members) and a high-touch customer support model tailored to the unique hours of combat sports gyms.

4. Future Vision: Advanced AI Integrations

Once the core SaaS infrastructure is stabilized, AI can transform the platform from a static database into an intelligent operational partner. Below are advanced AI feature concepts and their required engineering stacks.

A. The "Omniscient" Gym Owner Co-Pilot (CRM LLM)

Feature: A chat interface where the gym owner can ask complex analytical questions in natural language. (e.g., "Which members attended 3+ times last week but haven't paid this

month's invoice?", or "What's my projected MRR for next month based on historical churn?")

Engineering Stack:

- *Model:* OpenAI GPT-4o or Anthropic Claude 3.5 Sonnet.
- *Technique:* Text-to-SQL or Agentic RAG. Using frameworks like LangChain or LlamaIndex to translate natural language queries into secure SQL database queries, execute them on a read-replica of the PostgreSQL database, and return a synthesized answer.

B. Autonomous 24/7 Lead Conversion Agent

Feature: An AI agent plugged into the gym's Instagram DMs, website widget, and SMS. It acts as a virtual front-desk manager: answering pricing questions, overcoming objections, checking the live CRM schedule, and directly booking introductory classes for leads.

Engineering Stack:

- *Architecture:* Webhooks to capture inbound messages (Meta Graph API, Twilio).
- *AI Stack:* LLM with function-calling capabilities (OpenAI API) to read/write directly to the CRM's scheduling API endpoints.
- *Guardrails:* Semantic routing (e.g., using a fast embedding model) to detect if a conversation requires human intervention.

C. Computer Vision for Fight Form & Sparring Analytics

Feature: Gyms can set up a smartphone or camera on a heavy bag or sparring ring. The AI analyzes the video feed to track metrics like punch volume, velocity, combinations thrown, and fatigue over rounds. This data syncs to the member's profile in the CRM, giving coaches objective metrics.

Engineering Stack:

- *Edge/Cloud Vision:* OpenCV combined with pose estimation models (e.g., MediaPipe or YOLOv8-Pose).
- *Processing:* If processing in the cloud, utilizing AWS Rekognition or dedicated GPU instances running specialized PyTorch action-recognition models.
- *Data Sync:* Time-series database integration to log punch metrics per session.

D. Predictive Churn & Injury Modeling

Feature: An advanced machine learning model that acts as a radar for gym health. It looks at attendance decay, late payments, and training frequency. It flags members with a "High Risk of Canceling" score before they actually quit. For the "Fight Camp" module, it can detect overtraining (e.g., sparring volume vs. rest days) and warn coaches of high injury risk.

Engineering Stack:

- *Models:* This does not require an LLM. It relies on classical Machine Learning (XGBoost, Random Forest, or deep neural networks for time-series analysis like LSTMs).
- *Pipeline:* Scikit-learn or TensorFlow. Scheduled batch-processing jobs (e.g., via Apache Airflow) that run nightly over the CRM data, calculate risk scores, and update the owner's dashboard in the morning.